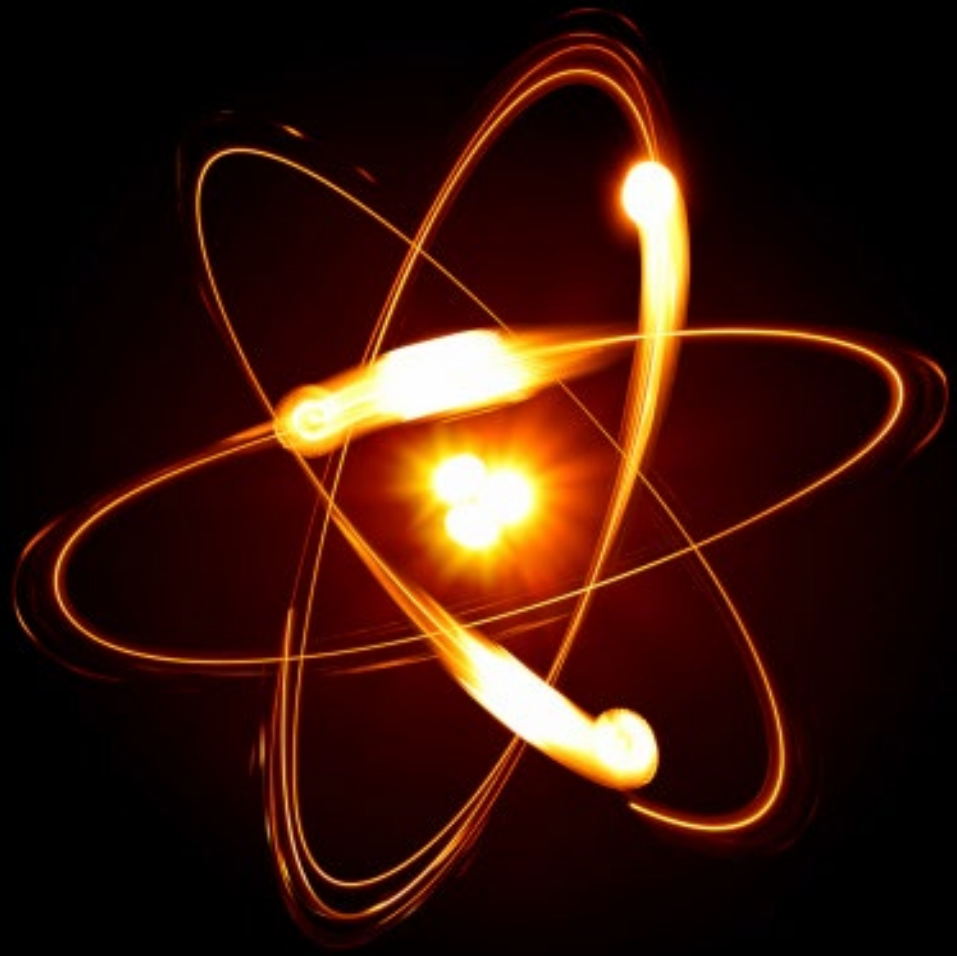
# Chapter 24.

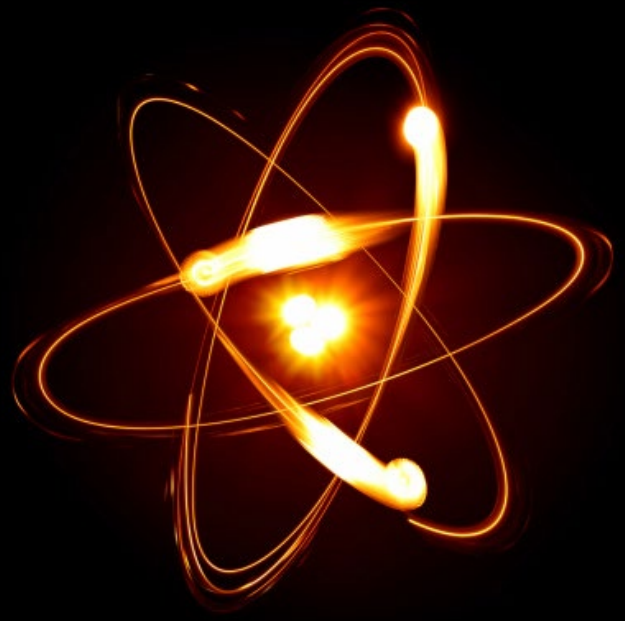
# Many-electron Atoms

# Part 1



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Quantum Physics



By convention sweet is sweet, bitter is bitter, hot is hot, color is color; but in truth there are only atoms and the void.

Democritus of Abdera
(c 400 BC)

# Chapter 24.

# Many-electron Atoms. Part 1

- The Many-Electron Atom: The Physical Problem
- The Many-Electron Hamiltonian
- First Approximations: Removing Electron–Electron Complexity
- Effective Potentials and Central Field Approximation
- Hartree and Hartree–Fock Models
- Ground State and Electronic Configuration
- The Periodic System and Atomic Trends

Bibliography

Eisberg and Resnick; B H Bransden and C J Joachain, Physics of Atoms and Molecules, Prentice Hall, 2003 ; Alonso and Finn (Vol. 3) ; H A Bethe and R Jackiw, Intermediate Quantum Mechanics, Addison Wesley, 1997

# Multi-electron atoms

The system is made up of a nucleus of atomic number $Z$ and $N \geq 2$ electrons ($N = Z$ for a neutral atom, $N \neq Z$ for an ion).

A detailed treatment should take into account

- Kinetic energy of the electrons and their potential energy in the electrostatic coulombic field created by the nucleus (assumed point-like and with infinite mass).
- Electrostatic coulombic repulsion between the electrons
- Magnetic interactions of the electronic spins with their orbital motion (spin-orbit interaction)
- Small effects (spin-spin interactions, relativistic effects, nuclear corrections, etc

In this Part 1, we'll take into account the first two points.

# Multi-electron atoms

The system is made up of a nucleus of atomic number $Z$ and $N \geq 2$ electrons ($N = Z$ for a neutral atom, $N \neq Z$ for an ion).

Therefore, the system Hamiltonian is

$$\mathrm{H} = -\frac{\hbar^2}{2\mu}\sum_{\mathrm{i}=1}^{\mathrm{N}} \nabla_{\mathrm{i}}^2 - \mathrm{Z}\sum_{\mathrm{i}=1}^{\mathrm{N}} \frac{\mathrm{e}^2}{4\pi\epsilon_0 \mathrm{r}_{\mathrm{i}}} + \sum_{\substack{\mathrm{i,j} \\ \mathrm{i>j}}}^{\mathrm{N}} \frac{\mathrm{e}^2}{4\pi\epsilon_0 \mathrm{r}_{\mathrm{ij}}}$$

where $\mathrm{r}_{\mathrm{ij}} = \left|\vec{\mathrm{r}}_{\mathrm{i}} - \vec{\mathrm{r}}_{\mathrm{j}}\right|$ and we are neglecting the nuclear motion (mass polarization term), relativistic kinematic effects, the finite size of the nucleus, and magnetic effects (spin-orbit interactions)

We have a system of identical particles. The hamiltonian is spin-independent. Moreover, it must be invariant under an exchange of coordinates (this is the case).

We must solve the Schrödinger equation (ignore electronic spin, for the moment):

$$\mathrm{H}\Psi(\vec{\mathrm{r}}_1, \vec{\mathrm{r}}_2, \ldots, \vec{\mathrm{r}}_{\mathrm{N}}) = \mathrm{E}\Psi(\vec{\mathrm{r}}_1, \vec{\mathrm{r}}_2, \ldots, \vec{\mathrm{r}}_{\mathrm{N}})$$

# Multi-electron atoms

Since we are dealing with a system containing $N$ indistinguishable particles, its Hamiltonian must be invariant under an interchange of the coordinates (spatial and spin) of any two particles.

This is clearly the case for the Hamiltonian which is independent of the spins of the electrons, and is symmetric in their spatial variables.

. Moreover, because electrons have spin 1/2 and hence are fermions, the *Pauli exclusion principle* requires that *the total wave function* $\psi(\mathbf{r}_1, \mathbf{r}_2, \ldots \mathbf{r}_N)$ *be completely antisymmetric*, i.e. changes sign if the coordinates (spatial and spin) of any two electrons are interchanged.

Once the spatial eigenfunctions have been obtained, the full eigenfunctions of the system are products of the spatial ones times spin wave functions.

# Can it be solved analytically?

The previous SE equation is a partial differential eqn. in 3N dimensions. It is **not** separable. (because of the appearance of the terms $1/r_{ij}$) Solve the previous SE equation? **Nope**

Even for the simplest case, ($N$ = 2, that is, He atom, or the ions $H^-$, $Li^+$, etc.) this equation **cannot be solved analytically** (because it is not an independent-particle hamiltonian).

Therefore we must

i. use numerical methods to solve the equation,

and/or

ii. use approximations in order to try and extract some information (semi)analytically

# (a) Approximations to the Hamiltonian

**Approximation 1:** Ignoring the interaction among electrons (Independent particle model). Each electron moves independently of the others.

$$H = -\frac{\hbar^2}{2\mu}\sum_{i=1}^{N}\nabla_i^2 - Z\sum_{i=1}^{N}\frac{e^2}{4\pi\epsilon_0 r_i} = h(1) + h(2) + \cdots + h(N)$$

It is then a problem of independent identical particles, with a single-particle hamiltonian given by that of a hydrogen-like atom of nuclear charge *Z*, whose energies and eigenfunctions are well known.

If we write

$$H\Psi_c(\vec{r}_1, \vec{r}_2, \ldots, \vec{r}_N) = \sum_{i=1}^{N}\left[-\frac{\hbar^2}{2\mu}\nabla_i^2 - Z\frac{e^2}{4\pi\epsilon_0 r_i}\right]\Psi_c(\vec{r}_1, \vec{r}_2, \ldots, \vec{r}_N) = E\Psi_c(\vec{r}_1, \vec{r}_2, \ldots, \vec{r}_N)$$

This is separable into *N* eqns, one for each electron.

A solution for $\Psi_c(\vec{r}_1, \vec{r}_2, \ldots, \vec{r}_N)$ can be written in terms of $\varphi_{\alpha_1}(\vec{r}_1), \varphi_{\alpha_2}(\vec{r}_2), \ldots, \varphi_{\alpha_N}(\vec{r}_N)$ where the individual electronic orbitals are solutions of, $\alpha_i = (n_i, l_i, m_i)$

$$\left[-\frac{\hbar^2}{2\mu}\nabla_i^2 - Z\frac{e^2}{4\pi\epsilon_0 r_i}\right]\varphi_{n_i,l_i,m_i}(\vec{r}_i) = e_{n_i}\varphi_{n_i,l_i,m_i}(\vec{r}_i)$$

# (a) Approximations to the Hamiltonian

where ( $E_l$ = 13.6 eV)

$$e_{n_i} = e_n = {}^{-E_l Z^2}/_{n^2} \qquad \text{degeneracy: } 2n^2$$

$$2\sum_{l=0}^{n-1}(2l+1) = 2\left(\frac{1+2n-1}{2}\right)n = 2n^2$$

If we include the electronic spin (we represent it by the ket $|\chi_{1/2,m_s}>$)

$$|\psi\rangle = |\varphi\rangle \otimes \left|\chi_{1/2,m_s}\right\rangle = \left|nlm\chi_{1/2,m_s}\right\rangle$$

$$n = 1,2,\ldots \;; \quad l = 0,1,\ldots,n-1; \quad m = -l,\ldots,l; \quad m_s = -\frac{1}{2},\frac{1}{2}$$

The quantum state and total energy of the atom are obtained by filling the hydrogen-like single-particle levels with electrons, in order of increasing energy, and taking into account Pauli's exclusion principle.

# He atom without e-e repulsion

The simplest case is that of atoms with **two electrons**, $H^-$(Z=1), He(Z=2), $Li^+$(Z=3). Now

$$H = -\frac{\hbar^2}{2\mu}\nabla_1^2 - \frac{\hbar^2}{2\mu}\nabla_2^2 - Z\frac{e^2}{4\pi\epsilon_0 r_1} - Z\frac{e^2}{4\pi\epsilon_0 r_2} + \frac{e^2}{4\pi\epsilon_0 r_{12}}$$

First, we discard the final term (independent particle model). We have $H = h(1) + h(2)$

$$h(i)\varphi_{n_i,l_i,m_i}(\vec{r}_i) = e_{n_i}\varphi_{n_i,l_i,m_i}(\vec{r}_i)$$

and $e_{n_i} = e_n = -E_l Z^2/n^2 \quad (E_l = 13.6\ eV)$

Consider He (Z = 2). Both electrons are in the ground state. Then, the total energy would be $2e_1$.

For one electron, its ground state energy will be $e_1$

$$e_1 = -Z^2\, E_l/n^2 = -2^2\frac{13.6}{1^1} = -54.4\ eV$$

# He atom without e-e repulsion

A solution for the two-particle hamiltonian would be

$$\Psi(\vec{r}_1, \vec{r}_2) = \varphi_{n_1,l_1,m_1}(\vec{r}_1)\varphi_{n_2,l_2,m_2}(\vec{r}_2)$$

$$E_{n_1,n_2} = e_1 + e_2$$

BUT, we have identical particles, so.....

Apply the symmetrizer and antisymmetrizer operators.

Two posibilities:

$$\Psi(\vec{r}_1, \vec{r}_2) = \frac{1}{\sqrt{2}}\left[\varphi_{n_1,l_1,m_1}(\vec{r}_1)\varphi_{n_2,l_2,m_2}(\vec{r}_2) \pm \varphi_{n_1,l_1,m_1}(\vec{r}_2)\varphi_{n_2,l_2,m_2}(\vec{r}_1)\right]$$

In the ground state $n_1 = n_2 = 1\,,\quad l_1 = l_2 = 0\,,\quad m_1 = m_2 = 0$. So

$$\Psi(\vec{r}_1, \vec{r}_2) = \varphi_{1,0,0}(\vec{r}_1)\varphi_{1,0,0}(\vec{r}_2)$$

The ground state energy of the He atom would be

$$2e_1 = 2(-2^2\, E_l/1) = -8E_l \approx -108.8\; eV$$

# He atom without e-e repulsion

BUT.....

$$\Psi(\vec{r}_1,\vec{r}_2)=\varphi_{1,0,0}(\vec{r}_1)\varphi_{1,0,0}(\vec{r}_2)$$

is symmetric and **we know** that....

the total wave function must be anti-symmetric. Therefore, for the spin contribution we must take the singlet state

$$\frac{1}{\sqrt{2}}\{|+,-\rangle-|-,+\rangle\}$$

and then we have .....

$$\{\varphi_{1,0,0}(\vec{r}_1)\varphi_{1,0,0}(\vec{r}_2)\}\otimes\frac{1}{\sqrt{2}}\{|+,-\rangle-|-,+\rangle\}=SLATER(|1;100+\rangle,|2;100-\rangle)=$$

$$=\frac{1}{\sqrt{2!}}\begin{vmatrix}\varphi_{100}(1)|+\rangle & \varphi_{100}(1)|-\rangle\\ \varphi_{100}(2)|+\rangle & \varphi_{100}(2)|-\rangle\end{vmatrix}=\frac{1}{\sqrt{2!}}\begin{vmatrix}|1;100+\rangle & |1;100-\rangle\\ |2;100+\rangle & |2;100-\rangle\end{vmatrix}$$

Experimental ground energy: –78.98 eV ⇒ **Not a good approximation.**

(Remember that we have neglected the e-e interaction which should give a positive contribution)

# He atom without e-e repulsion

For the **first excited** state ($1s^1 2s^1$) we obtain

$$E = e_{1s} + e_{2s} = -4\ {}^{E_l}/_{1} - 4\ {}^{E_l}/_{4} = -5E_l \approx -68.8\ eV\ \ (\text{Exp.} \approx -62.0\ eV)$$

The wave function: we have $\varphi_{1,0,0}$ and $\varphi_{2,l,m}$. Two possibilities:

$$\frac{1}{\sqrt{2}}\left[\varphi_{1,0,0}(\vec{r}_1)\varphi_{2,l,m}(\vec{r}_2) + \varphi_{2,l,m}(\vec{r}_1)\varphi_{1,0,0}(\vec{r}_2)\right]\otimes\frac{1}{\sqrt{2}}\{|+,-\rangle - |-,+\rangle\}$$

OR

$$\frac{1}{\sqrt{2}}\left[\varphi_{1,0,0}(\vec{r}_1)\varphi_{2,l,m}(\vec{r}_2) - \varphi_{2,l,m}(\vec{r}_1)\varphi_{1,0,0}(\vec{r}_2)\right]\otimes\frac{1}{\sqrt{2}}\{|+,-\rangle + |-,+\rangle\}$$

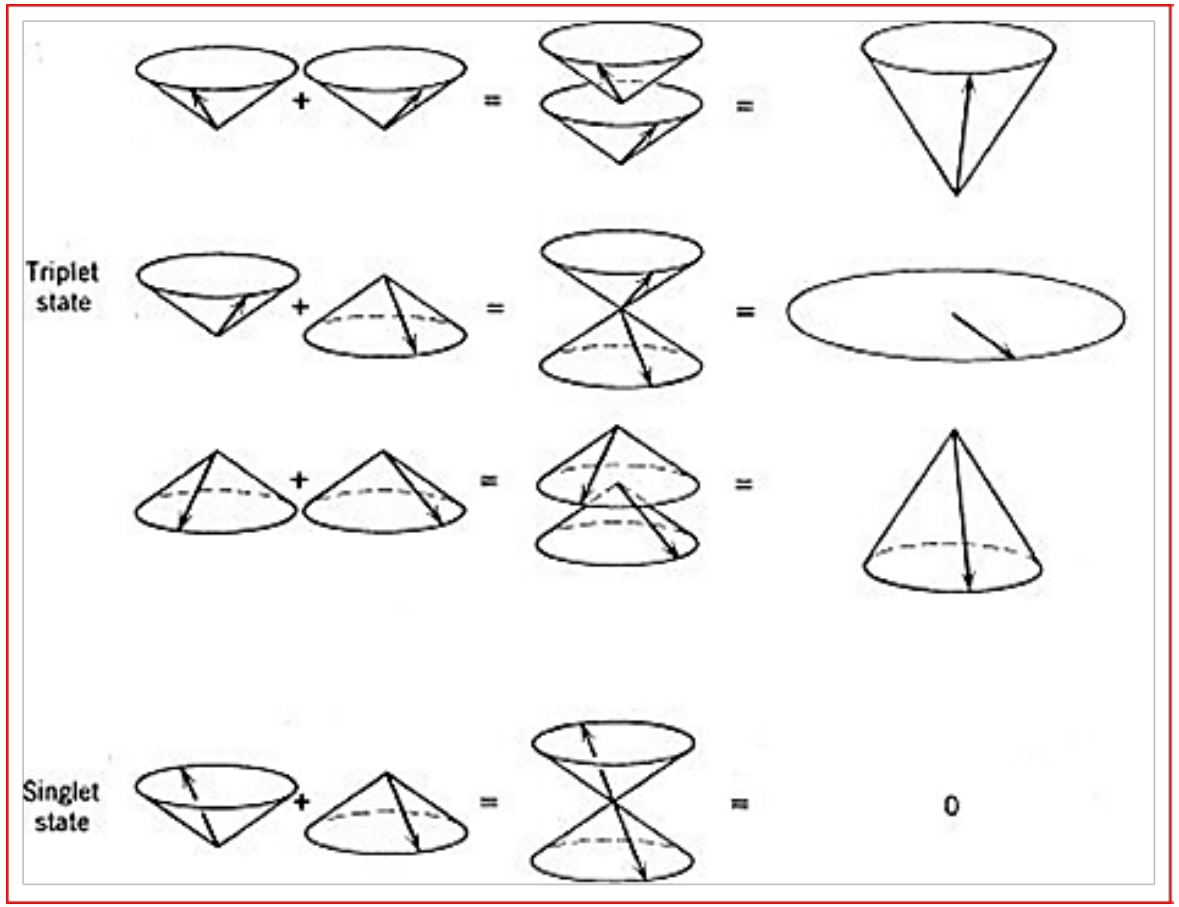
$$\frac{1}{\sqrt{2}}\left[\varphi_{1,0,0}(\vec{r}_1)\varphi_{2,l,m}(\vec{r}_2) - \varphi_{2,l,m}(\vec{r}_1)\varphi_{1,0,0}(\vec{r}_2)\right]\otimes|+,+\rangle \qquad \frac{1}{\sqrt{2}}\left[\varphi_{1,0,0}(\vec{r}_1)\varphi_{2,l,m}(\vec{r}_2) - \varphi_{2,l,m}(\vec{r}_1)\varphi_{1,0,0}(\vec{r}_2)\right]\otimes|-,-\rangle$$

Another interesting magnitude is the ionization energy (from the ground state).

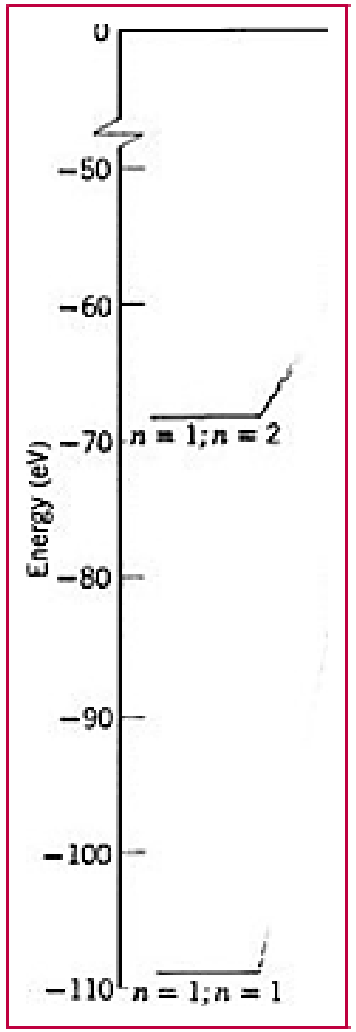
This would be $E_{ionizacion} = e_1 - 2e_1 \approx$ 54.4 eV, compared to an experimental value of 24.6 eV.

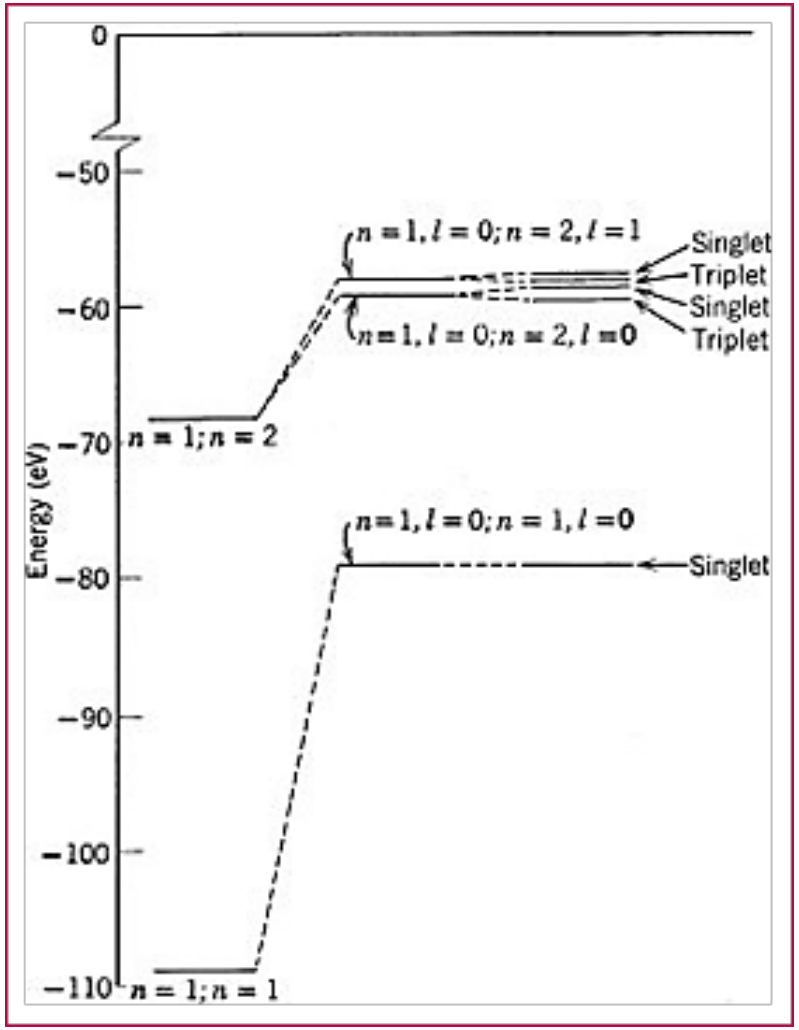
# He atom without e-e repulsion

Singlet and triplet states of a two-electron system



# He atom without e-e repulsion



# Can we improve this result? Perturbation approach

Because the ground state ($1s^2$) is a singlet state, the wave function (spatial part) must be symmetric

$$\Psi(\vec{r}_1, \vec{r}_2) = \phi_{100}(\vec{r}_1)\phi_{100}(\vec{r}_2)$$

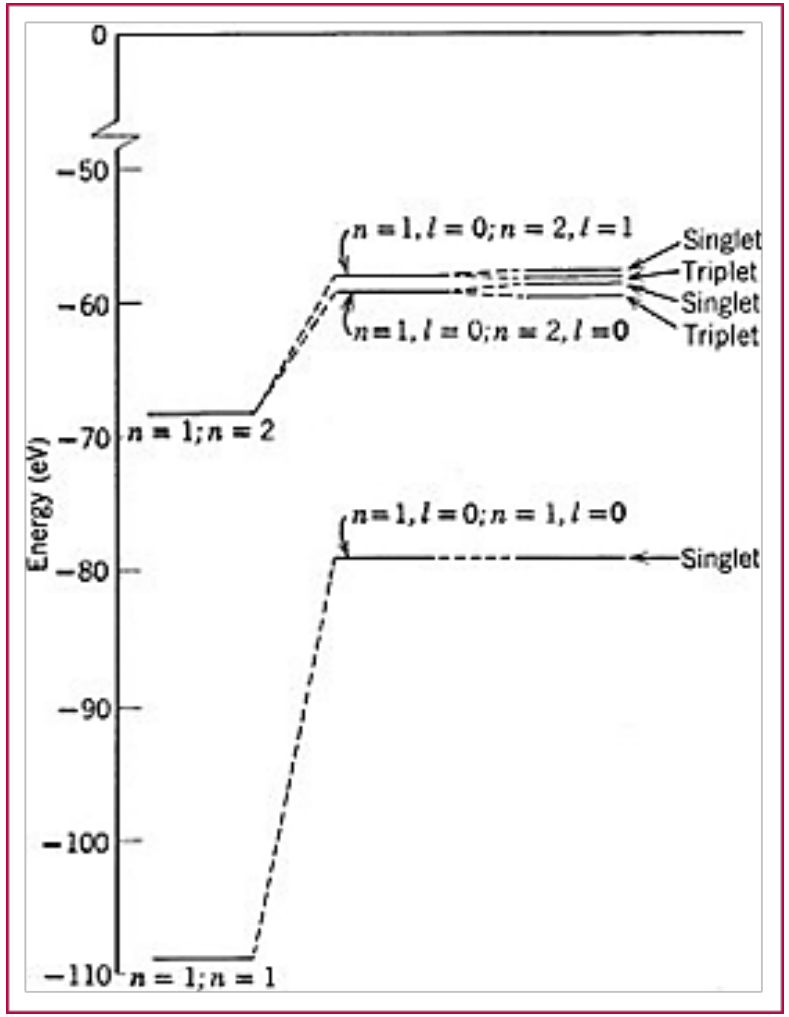
Assuming that the wave function itself is not affected by the electron repulsion, we can calculate that contribution (its mean value) to the energy:

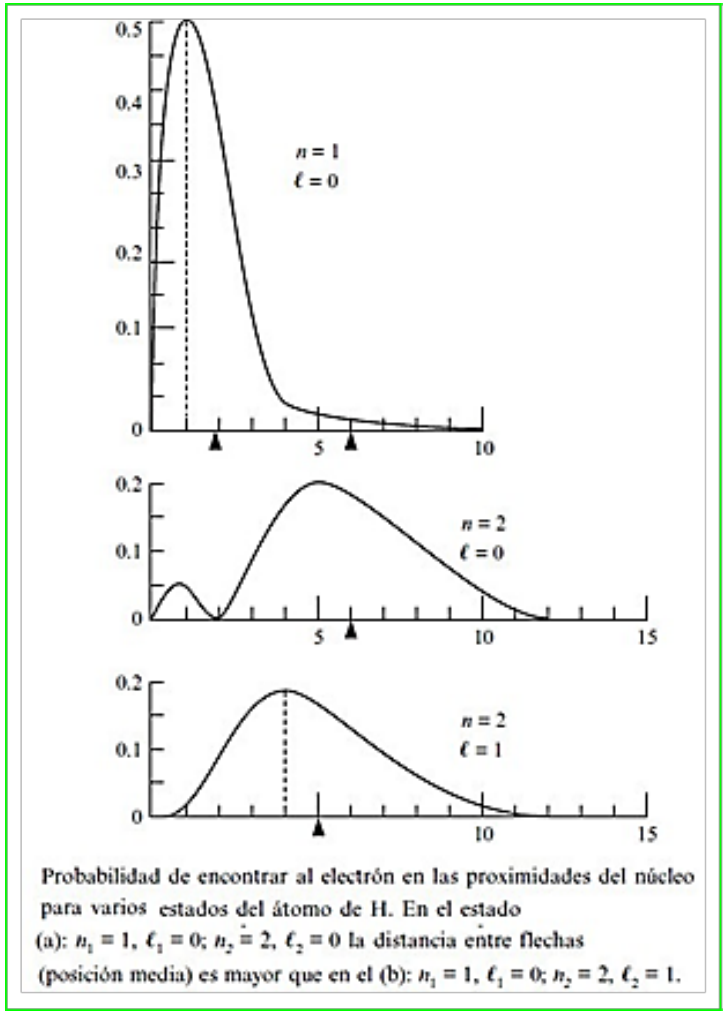
$$< E_{ee} > = \int d^3\vec{r}_1 d^3\vec{r}_2 \, |\phi_{100}(\vec{r}_1)|^2 \frac{e^2}{|\vec{r}_1 - \vec{r}_2|} |\phi_{100}(\vec{r}_2)|^2 = \frac{5Z}{4} E_l = +34 \, eV$$

so that the total energy would be $E(1s^2) = -108.8 + 34 = -74.8 \, eV$, which compares much better with the experimental value of -78.98.

Nevertheless, the error is still sizeable! **because we did not account for the screening** (and this error will increase with Z)

# He atom without e-e repulsion




Probabilidad de encontrar al electrón en las proximidades del núcleo para varios estados del átomo de H. En el estado (a): $n_1 = 1$, $\ell_1 = 0$; $n_2 = 2$, $\ell_2 = 0$ la distancia entre flechas (posición media) es mayor que en el (b): $n_1 = 1$, $\ell_1 = 0$; $n_2 = 2$, $\ell_2 = 1$.

# Can we improve this result? Screening .

**SCREENING**

An electron moves in a field created by the nucleus plus that of the other electron.

We may take that into account by assuming that the electron moves in an **effective** field created by a smaller nuclear charge.

So, replace $Z$ for $(Z - \delta)$

Now, the ground state energy would be written as

$$2e_1 = 2(-(Z-\delta)^2 \; E_l/1^2)$$

If we take $\delta \approx 0.32$, we get a good agreement with experiment.

# (b) Approximations to the ground state

**Variational method.**

This method is applied to the *evaluation of the ground state energy* of a system, provided we know its hamiltonian. The idea is to start with an approximation for the ground state wave function (which depends on one or several parameters). Then, evaluate $\langle\psi|H|\psi\rangle$ and minimize versus those parameters.

Its usefulness depends strongly on our ability to make a good initial guess for the form of the ground state eigenfunction.

Given a Hamiltonian *H* and a normalized state $|\psi\rangle$ the following inequality is verified:

$$E_g \leq \langle\psi|H|\psi\rangle$$

where $E_g$ is the exact ground state energy for *H*, and the equality holds only if $|\psi\rangle$ is the ground state.

If the state is not normalized, then the inequality reads $E_g \leq \frac{\langle\psi|H|\psi\rangle}{\langle\psi|\psi\rangle}$

The practical application starts with a family of kets $|\psi(\alpha)\rangle$ ,which depend on a parameter $\alpha$. Then evaluate $\langle H\rangle(\alpha)$ and minimize with respect to $\alpha$.

# Atomic units

For the hydrogen atom (Z=1), we obtained that its first Bohr radius was

$$a_0 = \frac{(4\pi\epsilon_0)\hbar^2}{m_e Z e^2} = 0.529 \text{ Å}$$

In atomic physics, it has been found convenient to introduce a new set of units (a.u.) in which $a_0$ is taken as the unit of length.

Consequently, $m_e$ is the unit of mass, $\hbar$ the unit of angular momentum, $e$ is the unit of charge and the permitivity of free space is taken as $\epsilon_0 = 1/4\pi$.

In conclusion ( $m = \hbar = e = 1$, $4\pi\epsilon_0 = 1$)

Now, for the energy (remember that $E_n = -\frac{1}{2}\left(\frac{Ze^2}{4\pi\epsilon_0}\right)\frac{m_e}{n^2\hbar^2}$

So, when written in a.u. : $E_n = -\frac{1}{2}\frac{Z^2}{n^2}$

but for Z=1 and n=1, we have $E_1 = -\frac{1}{2}$ a.u. and we know that $E$ = -13.6 eV.

1 *a.u.* = 27.2 *eV*

# Example of variational calculation: He atom

Apply the variational method to estimate the ground state energy of He. We choose $\psi(\vec{r}_1, \vec{r}_2)$ as a product of hydrogen-like functions in the 1*s* state.

The ground state wavefunction is (in a.u. units)

$$\psi(\vec{r}_1, \vec{r}_2) = \varphi_{100}(r_1)\,\varphi_{100}(r_2) = \frac{1}{\pi}\left(\frac{2}{1}\right)^3 e^{-2(r_1+r_2)}$$

The 2–factor corresponds to the nuclear charge $Z = 2$ of the He atom. However, the electronic screening suggests that it could be more adequate to use a smaller value of $Z$ in the expression of the wavefunction. **(not in the Hamiltonian)**.

So we take a trial wavefunction depending on one parameter ($Z^*$)

$$\psi(Z^*, \vec{r}_1, \vec{r}_2) = \frac{1}{\pi}\left(\frac{Z^*}{1}\right)^3 e^{-Z^*(r_1+r_2)}$$

then compute $\langle H \rangle = E(Z^*) = \int d\vec{r}_1 \int d\vec{r}_2\, \psi^*(Z^*; \vec{r}_1, \vec{r}_2)\, H\, \psi(Z^*; \vec{r}_1, \vec{r}_2)$

with $H$ being the **exact** Hamiltonian, and minimize $E(Z^*)$ with respect to $Z^*$.

The minimum value will be an approximation (from above) for $E_g$.

# Example of variational calculation: He atom

Remember that

$$H = -\frac{\hbar^2}{2\mu}\nabla_1^2 - -\frac{\hbar^2}{2\mu}\nabla_2^2 - Z\frac{e^2}{4\pi\epsilon_0 r_1} - Z\frac{e^2}{4\pi\epsilon_0 r_2} + \frac{e^2}{4\pi\epsilon_0 r_{12}}$$

So, we have to evaluate ( in a.u.)

$$\langle\psi(Z^*;\vec{r}_1,\vec{r}_2)|(H_1 + H_2 + 1/r_{12})|\psi(Z^*;\vec{r}_1,\vec{r}_2)\rangle$$

where

$$H_1 = T_1 - Z/r_1 \qquad H_2 = T_2 - Z/r_2 \qquad \langle\psi|H_1|\psi\rangle = \langle\psi|H_2|\psi\rangle = \frac{Z^*}{2}(Z^* - 2Z)$$

$$\left\langle\psi\left|\frac{1}{r_{12}}\right|\psi\right\rangle = \frac{5}{8}Z^* \qquad \langle\psi|H|\psi\rangle = E(Z^*) = (Z^*)^2 - 2ZZ^* + \frac{5}{8}Z^*$$

$$\frac{\partial E}{\partial Z^*} = 0 \qquad Z^* = Z - \frac{5}{16}$$

The result is (we'll solve it as an exercise) $E(Z_{min}) = -77.38\,eV$, to compare with the experimental one $(-78.98\,eV)$ which has an error around 2%

# Effective potentials

Remember that our problem was to solve a N-electrons hamiltonian

$$\left[-\frac{\hbar^2}{2\mu}\sum_{i=1}^{N}\nabla_i^2 - Z\sum_{i=1}^{N}\frac{e^2}{4\pi\epsilon_0 r_i} + \sum_{\substack{i,j \\ i>j}}^{N}\frac{e^2}{4\pi\epsilon_0 r_{ij}}\right]\Psi(\vec{r}_1,\vec{r}_2,\dots,\vec{r}_N) = E\,\Psi(\vec{r}_1,\vec{r}_2,\dots,\vec{r}_N)$$

We know that this is a partial differential eqn. in 3N dimensions.

It is **not** separable. (because of the terms $1/r_{ij}$ )

Solutions? ⇒ **Central field approximation** (Hartree, Hartree-Fock)

Based on the independent particle model, it assumes that each electron moves in an **effective potential** which represents the attraction of the nucleus and the *average effect* of the repulsive interactions with the other (N-1) electrons.

# Effective potentials

**Approximation 2a:** Construct **effective potentials** $V_i(\vec{r}_i)$ such that

$$-Z\sum_{i=1}^{N}\frac{e^2}{4\pi\epsilon_0 r_i}+\sum_{\substack{i,j\\ i>j}}^{N}\frac{e^2}{4\pi\epsilon_0 r_{ij}}=\sum_{i=1}^{N}V_i(\vec{r}_i)+W$$

and *W* is a "small" term, compared to the $V_i(\vec{r}_i)$ (which is not the case for the e-e repulsion), so that if it is neglected, the error will be small.

After neglecting *W* the hamiltonian will be

$$H=\sum_{i=1}^{N}h_i(i)=\sum_{i=1}^{N}\left[-\frac{\hbar^2}{2\mu}\nabla_i^2+V_i(\vec{r}_i)\right]$$

But, how to construct such $V_i(\vec{r}_i)$ ????

# Asymptotic behaviour of V(r)

$V_i(\vec{r}_i)$ represents an **effective potential** "felt" by electron $i$ at $\vec{r}_i$ due to the attraction of the nucleus plus the average effect of the repulsive interactions among this electron and the other (N-1) electrons.

Consider the *i*-electron located at $\vec{r}_i$ . Its potential energy, $V_i(\vec{r}_i)$ , will be (in a.u.)

$$V_i(\vec{r}_i) = -\frac{Z}{r_i} + \sum_{\substack{j=1 \\ j\neq i}}^{N} \frac{1}{r_{ij}}$$

1) If $r_i$ is large (in comparison to the other $r_j$) then we can approximate $r_{ij} \approx r_i$ and therefore

$$V_i(\vec{r}_i) = -\frac{Z}{r_i} + \sum_{j=1}^{N-1} \frac{1}{r_i} = -\frac{Z-(N-1)}{r_i}$$

This is the Coulomb field of the nucleus screened by the other (N - 1) electrons.

# Asymptotic behaviour of V(r)

2) As the distance $r_i$ decreases, the screening effect becomes less effective and when $r_i$ is close to the nucleus, then we can approximate $r_{ij} \approx r_j$ and therefore

$$V_i(\vec{r}_i) = -\frac{Z}{r_i} + \left\langle \sum_{j=1}^{N-1} \frac{1}{r_j} \right\rangle \approx -\frac{Z}{r_i} + Cte$$

where the brackets denote the average of the inverse distances of the other (N -1) electrons.

Therefore, the **effective potential** $V_i(\vec{r})$ is such that

$$V_i(\vec{r}) \approx -\frac{Z}{r} \qquad r \to 0$$

$$V_i(\vec{r}) \approx -\frac{Z-(N-1)}{r} \qquad r \to \infty \qquad \text{(for neutral atom } V_i(\vec{r}) = -\frac{1}{r} \text{ when } r \to \infty)$$

The determination of $V_i(\vec{r})$ for **intermediate distances** is much more complicated. **It depends on the charge distribution of the electrons.** There are different schemes: Hartree, Hartree-Fock, Thomas-Fermi etc.

# Effective potentials

**Approximation 2b:** Assume that all the **effective potentials** are equal (that is, they are independent of *i* ) .Then

$$H = \sum_{i=1}^{N} h(i) = \sum_{i=1}^{N} \left[ -\frac{\hbar^2}{2\mu} \nabla_i^2 + V(\vec{r}_i) \right]$$

We face again a problem of independent identical particles, and the single-particle hamiltonians are the same, although notice that they **are not those of a hydrogen-like atom.**

In fact, the effective potential may not even be spherically symmetric!

This effective potential includes the attraction of the nucleus and the average effect of the repulsive interactions due to the other electrons.

*Note: The single-particle eigenstates, from which the atomic eigenstate is formed, are called atomic orbitals.*

# Central field model

**Approximation 2c:** Assume that $V(\vec{r}_i)$ is spherically symmetric, or else perform a spherical average (over angles), so that

$$H = \sum_{i=1}^{N} h(i) = \sum_{i=1}^{N} \left[ -\frac{\hbar^2}{2\mu} \nabla_i^2 + V(r_i) \right]$$

The problem is now very similar to that of Approximation 1 (single-particle hamiltonians with spherically symmetric potentials) except that they are not hydrogen-like.

In general $V(r)$ is computed numerically, and the single-particle Schrödinger equation is also solved numerically to obtain the single-particle energies and orbitals.

Anyhow, some characteristics of the solutions can be obtained analytically.

# Properties of the orbitals and their energies.

As the potential $V(r_i)$ is spherically symmetric (central potential) then the one-electron orbitals will be products of radial functions times spherical harmonics.

A possible solution to the above hamiltonian (now, it is separable!) may be written as

$$\psi(\vec{r}_1, \dots, \vec{r}_N) = \phi_{\alpha_1}(\vec{r}_1)\,\phi_{\alpha_2}(\vec{r}_2)\dots\phi_{\alpha_N}(\vec{r}_N) \qquad \alpha_i = (n_i, l_i, m_i)$$

where the individual orbitals are solutions of

$$-\frac{1}{2}\nabla^2\phi_{n,l,m}(\vec{r}) + V(r)\phi_{n,l,m}(\vec{r}) = E_{n,l}\phi_{n,l,m}(\vec{r})$$

Taking into account that $V(r)$ is a central potential, then .....

# Properties of the orbitals and their energies.

Now, the one-electron orbitals are: $\phi_{n,l,m}(\vec{r}) = R_{n,l}(r)Y_{l,m}(\theta,\varphi)$

And the radial eqn. satisfies

$$-\frac{1}{2}\left(\frac{d^2}{dr^2}+\frac{2}{r}\frac{d}{dr}-\frac{l(l+1)}{r^2}\right)R_{n,l}(r)+V(r)R_{n,l}(r)=E_{n,l}R_{n,l}(r)$$

where $n = 1, 2, \ldots;$ $l = 0, 1, \ldots, n-1;$ $m = -l, -l+1, \ldots, +l$

- The $\phi_{n,l,m}$ are **not** the hydrogenic wave functions, because the $R_{n,l}(r)$ are solutions for a potential different from the Coulombic.
- $V(r)$ is spherically symmetric and the energy eigenvalues **do not** depend on $m$.
- Now, the energies depend on both $(n, l)$. The total energy of the atom is, in this approximation, the sum of the individual energies.
- We still have exchange degeneracy. We have to construct a total N-electron wave function which is antisymmetric in the exchange of the coordinates (spatial and spin) of any two electrons.

# Properties of the orbitals and their energies.

We have to introduce the spin of the electrons in our model

This is done by multiplying the one-electron spatial orbitals

$$\phi_{n,l,m}(\vec{r}) = R_{n,l}(r)Y_{l,m}(\theta,\varphi)$$

by a spin 1/2 eigenfunction $\chi_{\frac{1}{2},m_s}$.

Then, we have ($\vec{q}$ stands for spatial and spin coordinates)

$$\phi_{n,l,m,m_s}(\vec{q}) = \phi_{n,l,m}(\vec{r})\chi_{\frac{1}{2},m_s} = R_{n,l}(r)Y_{l,m}(\theta,\varphi)\chi_{\frac{1}{2},m_s}$$

Notice that the electronic state is characterized by four quantum numbers.

These spin-orbitals satisfy

$$-\frac{1}{2}\nabla^2\phi_{n,l,m,m_s}(\vec{q}) + V(r)\phi_{n,l,m,m_s}(\vec{q}) = E_{n,l}\phi_{n,l,m,m_s}(\vec{q})$$

They can also be specified as $|nlm\chi\rangle$ .Their space part will be represented by wavefunctions $R_{nl}(r)Y_{l,m}(\theta,\phi)$ and their spin part by the states $|+\rangle$ or $|-\rangle$.

# Properties of the orbitals and their energies.

The hamiltonian we are considering is spherically symmetric and spin independent. Therefore, the eigen-energies will not depend on $m$ or on $\chi$, so each individual electron energy level is $2(2\ell + 1)$ times degenerate.

Summarizing, we have

Orbitals: $|nlm\chi\rangle$ Energies: $e_{nl}$

*Note: The energies* **depend** *explicitly on $\ell$ contrary to the case of hydrogen-like atoms, that is, the energy of the level 2p is different from the one of the level 2s, and the same happens with the levels 3s, 3p and 3d, etc.*

# Properties of the orbitals and their energies.

Now, we have to construct an antisymmetric N-electron wavefunction

Let's refer to the set of the four quantum numbers ($n, l, m, m_s$) by the letters $\alpha$, $\beta$..... $\mu$

The total wave function $\Psi(\vec{q}_1, \vec{q}_2, \ldots, \vec{q}_N)$, describing an atom in which an electron is in state $\alpha$, another in state $\beta$ ...

$$\Psi = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_\alpha(\vec{q}_1) & \phi_\beta(\vec{q}_1) & \cdots & \phi_\mu(\vec{q}_1) \\ \phi_\alpha(\vec{q}_2) & \phi_\beta(\vec{q}_2) & \cdots & \phi_\mu(\vec{q}_2) \\ \vdots & \vdots & \cdots & \vdots \\ \phi_\alpha(\vec{q}_N) & \phi_\beta(\vec{q}_N) & \cdots & \phi_\mu(\vec{q}_N) \end{vmatrix}$$

It is a Slater determinant. The $\vec{q}_i$ represents the spatial and spin coordinates. $\phi_\gamma(\vec{q}_i) = \phi_{n,l,m}(\vec{r}_i)\chi_{1/2,m_s}$

This wave function is antisymmetric because if we interchange the (spatial and spin) coordinates of two electrons (say $q_1$ and $q_2$), this is equivalent to interchanging two rows, so the determinant changes sign.

Moreover, if two electrons have the same quantum numbers (i.e. $\alpha = \beta$ ) then two columns are equal and the determinant vanishes.

# But how is V (r ) constructed? Hartree model

Assume that each electron in a multielectron system, is described by its own wavefunction. Each electron moves in an effective potential due to the nucleus and the other electrons.

This latter contribution is obtained by postulating that there is a negative charge density associated to each electron which is (-e) times its position probability density.

Therefore, the potential for the *i* -th electron is

$$V_i(\vec{r}) = -\frac{Z}{r} + \sum_{\substack{j=1 \\ j\neq i}}^{N} \int d\vec{r}\,' \frac{\left|\phi_j(\vec{r}\,')\right|^2}{|\vec{r} - \vec{r}\,'|}$$

*j* represents a set of quantum numbers describing the state of the *j*-th electron whose position is given by $\vec{r}\,'$.

(We use *atomic units*)

If we have N electrons, this leads to a set of N simultaneous integro-differential eqns. of the form

$$-\frac{1}{2}\nabla^2\phi_i(\vec{r}) + V_i(\vec{r})\phi_i(\vec{r}) = e_i\phi_i(\vec{r})$$

# But how is V (r ) constructed? Hartree model

- Replace $\sum_{j=1,(j\neq i)}^{N}$ by $\sum_{j=1}^{N}$ . Now we write $V(\vec{r})$, as it becomes the same for all electrons. (notice the appearance of **self-interaction**)

- Replace $V(\vec{r})$ by its angular average

$$V(r) = \frac{1}{4\pi}\int V(\vec{r})d\Omega$$

- Finally, we end up with the following two equations

$$V(\vec{r}) = -\frac{Z}{r} + \sum_{j=1}^{N}\int d\vec{r}\,' \frac{|\phi_j(\vec{r}\,')|^2}{|\vec{r}-\vec{r}\,'|} \qquad -\frac{1}{2}\nabla^2\phi_i(\vec{r}) + V(r)\phi_i(\vec{r}) = e_i\phi_i(\vec{r})$$

to be solved together (self-consistently).

# But how is V (r ) constructed? Hartree model

*V* (*r* ) is the *averaged* spherically symmetric potential felt by an electron at *r* .

This is called the **HARTREE MODEL**:

**Summary**

Note that the total electronic density is $\rho(r) = \sum_j |\phi_j(r)|^2$ , where $\phi_j(r)$ are the atomic orbitals, obtained from the single-particle Schrödinger equation, and *j* runs over the occupied orbitals.

The atomic orbitals are obtained by solving the two (coupled) equations

$$V(\vec{r}) = -\frac{Z}{r} + \sum_{j=1}^{occ} \int d\vec{r}' \frac{|\phi_j(\vec{r}')|^2}{|\vec{r} - \vec{r}'|} \qquad \text{and} \qquad -\frac{1}{2}\nabla^2\phi_i(\vec{r}) + V(r)\phi_i(\vec{r}) = e_i\phi_i(\vec{r})$$

These two equations must be solved together (self-consistently).

# Hartree model. Where does it come from?

Hartree's model emerges as the result of imposing a "weak" version of the exclusion principle, where we take the trial wave function

$$\psi(\vec{r}_1, \dots, \vec{r}_N) = \phi_{\alpha_1}(\vec{r}_1) \dots \phi_{\alpha_N}(\vec{r}_N)$$

but at the same time we request that the orbitals are filled according to Pauli's exclusion principle.

Notice that the (total) wave function is NOT anti-symmetric (weak version)

We apply the variational principle to the previous trial wave function. The idea is to choose those one-electron orbitals $\phi_i$ that minimize $\langle H \rangle$

$$\langle H \rangle_\psi = \sum_{i=1}^{N} \int d\vec{r}\, \phi_i^*(\vec{r}) \left( -\frac{1}{2}\nabla^2 - \frac{Z}{r} \right) \phi_i(\vec{r}) + \sum_{i=1}^{N} \sum_{j>i} \int \int d\vec{r}\, d\vec{r}\,' \frac{|\phi_i(\vec{r})|^2 |\phi_i(\vec{r}\,')|^2}{|\vec{r} - \vec{r}\,'|}$$

# Hartree model. Practical implementation

Initially, the exact form of $V(r)$ is not known, but it can be found by a self-consistent procedure comprising several steps.

- An initial guess for $V(r)$ is given. It could be something like $V(r) = -Z/r$ for $r \to 0$ and $V(r) = -1/r$ for $r \to \infty$ with an educated guess for intermediate $r$-values.

- Now solve SE for the previous potential. The radial part must be solved numerically. The eigenfunctions are $\phi_\alpha(r,\theta,\varphi), \phi_\beta(r,\theta,\varphi), \phi_\gamma(r,\theta,\varphi)$... listed in order of increasing energies $E_\alpha < E_\beta < E_\gamma < \ldots$

  $\alpha, \beta, \gamma, \ldots$ refer to a set of three spatial and one spin quantum numbers

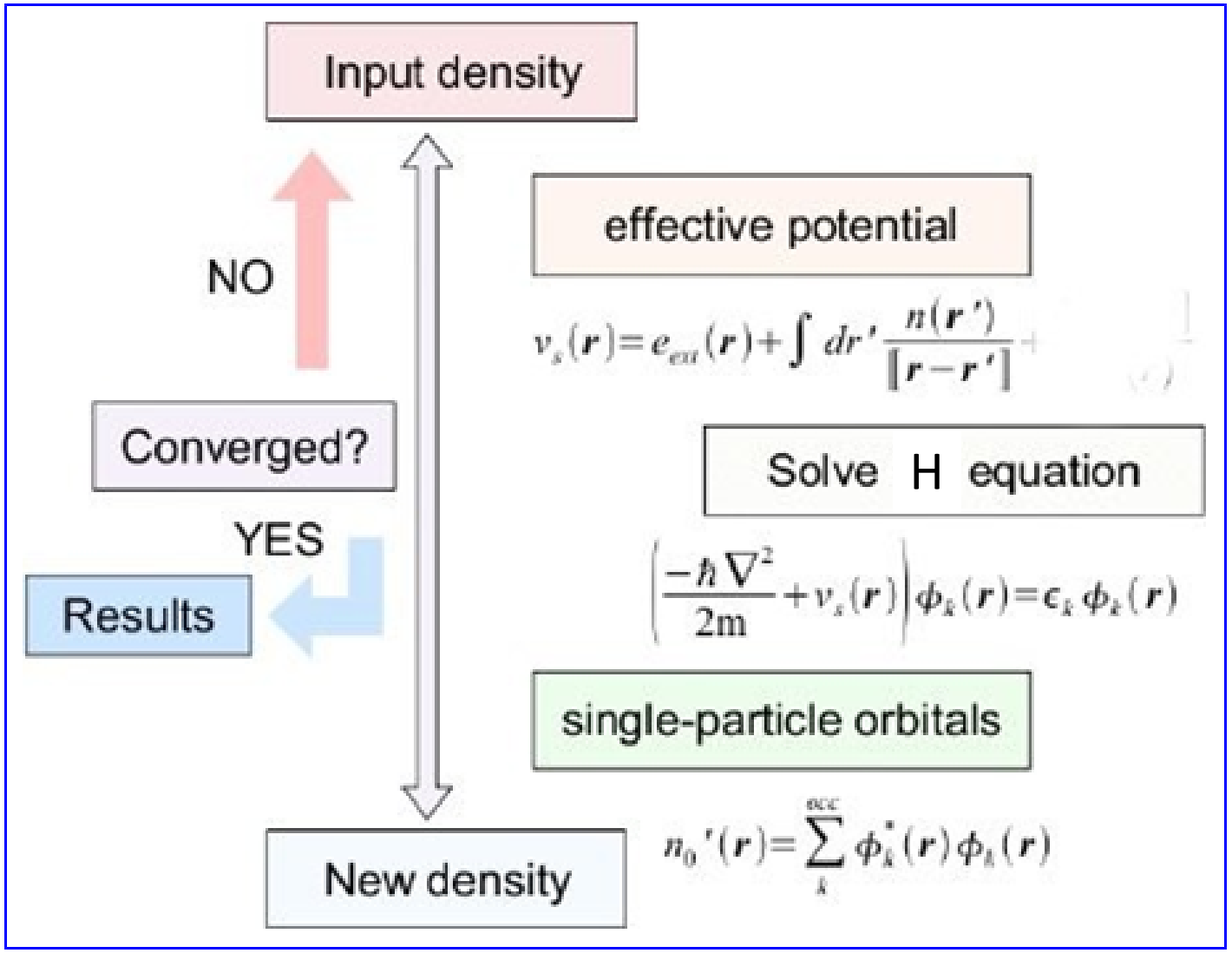
- The ground state energy of the atom is obtained by filling the electronic quantum states in such a way that the total energy is minimized. The states are filled in order of increasing energy and **one electron per state.**

# Hartree model. Practical implementation

- Now, the electronic charge densities are obtained from the previous eigenfunctions $\left(\rho(r) = \sum_j^{occ} \left|\phi_j(\vec{r})\right|^2\right)$
- Gauss's law of electrostatics is used to calculate the electric field produced by the total charge distribution of the previous step. From the electric field (by integration) a new estimate for the potential $V(r)$ is obtained.

  The $V(r)$ can also be obtained directly from the $\rho(r)$
- Now, go back to step 2 again.
- The whole process is repeated until self-consistency is achieved.
- The exclusion principle is satisfied because only one electron is allocated to each quantum state.
- However, the total wavefunction is **not** antisymmetric. It would be much more complicated to use a Slater determinant.

# Hartree self-consistent cycle



# Hartree model. Results

- The eigenfunctions are written as

$$\phi_{n,l,m,m_s}(r,\theta,\varphi) = R_{n,l}(r)Y_l^m(\theta,\varphi)\chi_{1/2,m_s}$$

- The *r*-dependence of $R_{n,l}(r)$ is **not** the same as that of a one electron atom.
- All electrons with the same *n* constitute a SHELL. Within a shell, the electrons with the same (*n*, *l* ) constitute a SUBSHELL.
- The sum of the probability densities for fixed *n* and *l* and **all** possible *m* values, is spherically symmetrical (and the same happens to the associated charge distribution). (Remember that $\sum_{m=-l}^{l}|Y_l^m(\theta,\varphi)|^2 = \frac{2l+1}{4\pi}$
- Therefore, when a multielectron atom is in its ground state, the lowest energy states are completely filled, which implies that all *m*-values are occupied. For those states, as the sum of the probability densities is spherically symmetrical, the same happens with the associated charge distribution. There may be a few electrons in the highest energy states (where not all possible *m*-values are filled) which can contribute with some asymmetries to the charge distribution.

# Hartree model. Results

We know that

$$\phi_{n,l,m}(r,\theta,\varphi) = R_{n,l}(r)Y_l^m(\theta,\varphi)$$

Probability of finding an (*n*, *l* )-electron inside the volume element $d\vec{r} = r^2 \sin\theta \, dr \, d\theta \, d\varphi$ **is proportional to** (the factor 2 takes spin into account)

$$2\sum_{m=-l}^{l} \left|\phi_{n,l,m}(r,\theta,\varphi)\right|^2 r^2 \sin\theta \, dr \, d\theta d\varphi$$

Taking into account that $\sum_{m=-l}^{l} |Y_l^m(\theta,\varphi)|^2 = \frac{2l+1}{4\pi}$ , we have

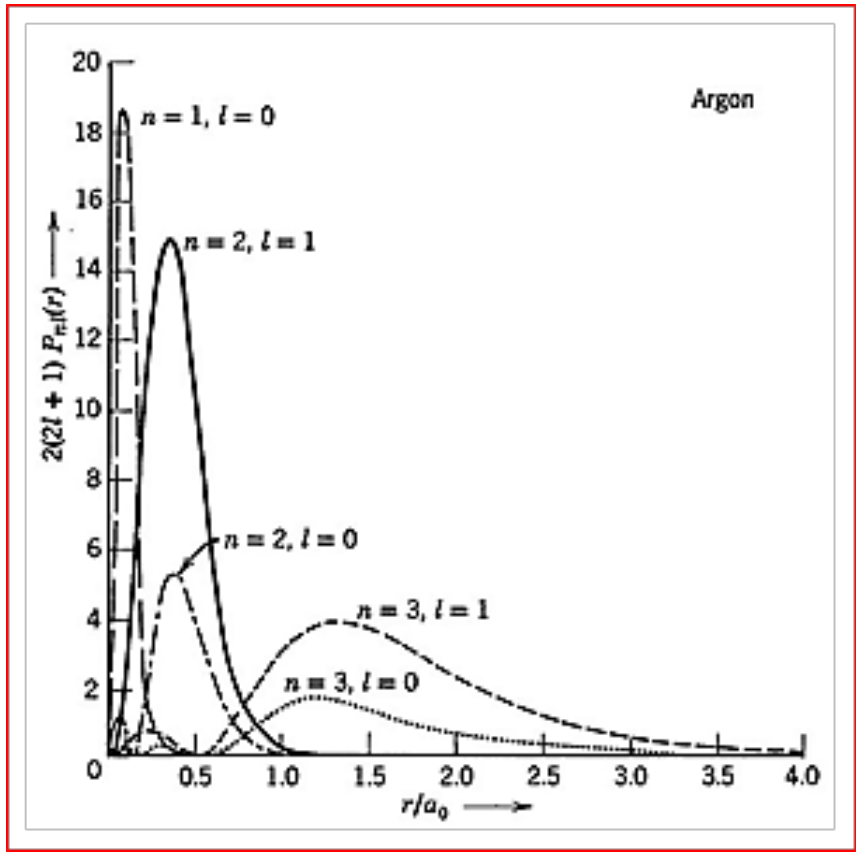
$$2\sum_{m=-l}^{l} \left|\phi_{n,l,m}(r,\theta,\varphi)\right|^2 r^2 \sin\theta \, dr \, d\theta \, d\varphi = 2\,\frac{2l+1}{4\pi} R_{n,l}^2(r) r^2 \sin\theta \, dr \, d\theta \, d\varphi$$

Evaluating the angular integrals, we have the probability of finding an (*n*, *l* )-electron within the range $(r, r+dr)$ **is proportional to**

$$2(2l+1)R_{n,l}^2(r)r^2 dr$$

# Hartree model. Results

Remember that $2(2l+1)r^2R_{n,l}^2(r)dr = 2(2l+1)P_{nl}(r)dr$ **is proportional to** the probability of finding an $(n,l)$ −electron within the range $(r, r+dr)$. The plot of $P_{nl}(r)$ shows that the curves are large within a small range. It is said that these electrons are in the same **subshell**. It is possible to define an $\bar{r}_{n,l}$ for each subshell.




Ar atom $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

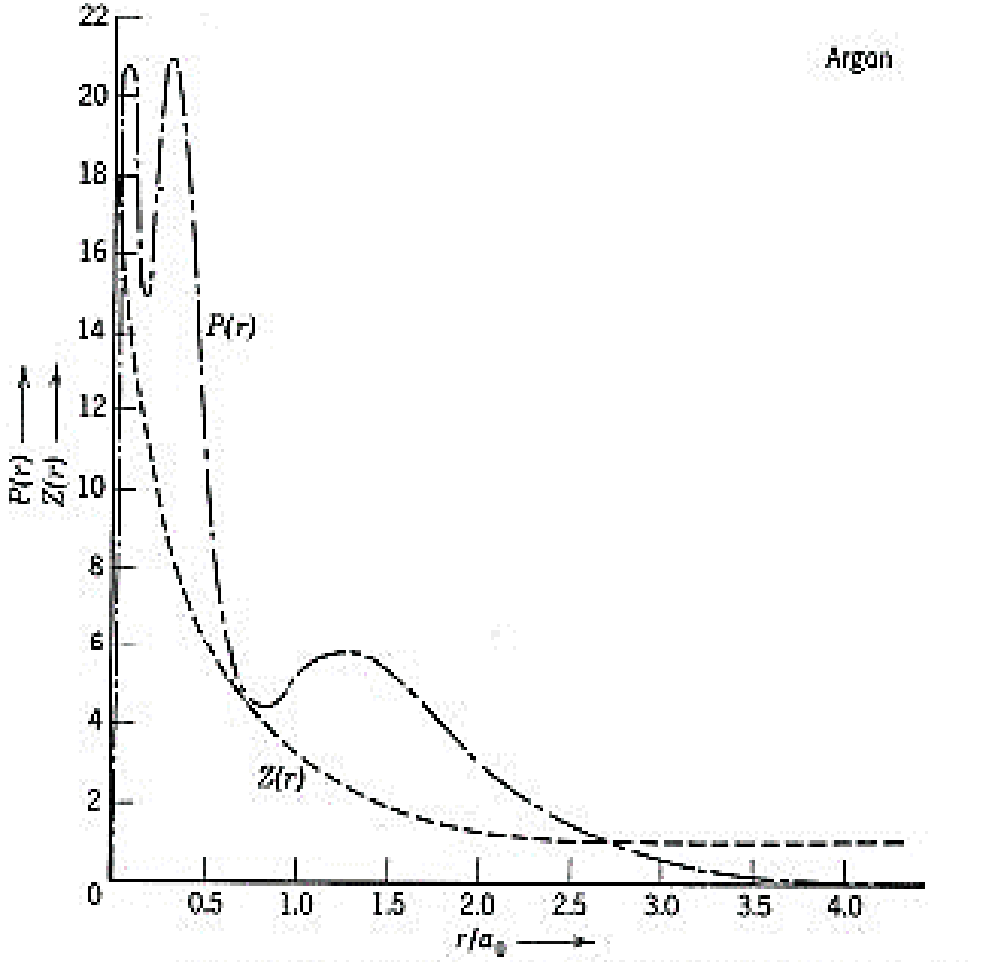
Plot of the radial probability density $2(2l+1)P_{nl}(r) = 2(2l+1)r^2R_{n,l}^2(r)$ as a function $r/a_0$.

# Hartree model. Results

**Summary**

- The sum of the probability densities for fixed ($n$, $l$) and all possible $m$ values is spherically symmetrical.
- When a multielectron atom is in its ground state, the lowest energy quantum states are completely filled, i.e. all possible $m$ values are occupied. As the sum of the probability densities is spherically symmetrical, the same happens to the charge distribution.
- Notice that the mean radius of the outermost shell is slightly larger than $a_0$, which is that of the hydrogen atom.
- On the other hand, the inner shell (n=1) has a very small radius.
- Total radial probability density $P(r) = \sum_{n,l} 2(2l+1)P_{n,l}(r)$. $P(r)dr$ is the probability of finding electrons within the range $r, r+dr$

# Hartree model. Results



The total radial probability density $P(r)$ of the argon atom

Plot of the total radial probability density.

Write $V(r) = -Z(r)/r$

$V(r)$ is the net potential as obtained from Hartree-type calculations.

# Hartree model. Results

- The calculation produces a $V(r)$ which increases from $-Z/r$ to $-1/r$.

  So we can write $V(r) = -Z(r)/r$ and assign a Z-value for each shell, $Z_n = Z(\bar{r}_n)$ (something like an effective Z for the shell n and $\bar{r}_n$ is the mean value of the radial coordinate for the shell)

  Example Ar atom $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

  From the graphs, we obtain $Z_1 \approx 16, Z_2 \approx 8, Z_3 \approx 3$.

- **Simple descriptive model**: The electrons in shell *n* are moving under a Coulombic potential $V_n(r) = -Z_n/r$. Use Bohr's one-electron eqns. for energies and radii

  Remember $r_n = n^2 a_0/Z$ , $E_n = (Z/n)^2(-13.6\ eV)$

- If we use the one-electron atom formula for the energy, but with the $Z_n$ , then we obtain an estimate for the energy of the electrons in each shell

  $E_1 \approx -(16/1)^2\ 13.6 = -3500\ eV$ $E_2 \approx -(8/2)^2\ 13.6 = -220\ eV$ $E_3 \approx -(3/3)^2\ 13.6 = -14\ eV$

  Errors of $\approx 20\%$

  As a rule, for multielectron atoms, $Z_1 \approx Z - 2$ and for the outermost shell $Z_n \approx n$

# Hartree model. Results

- The inner shells of small n have very small radii. Using Bohr's model, i.e. $r_n = n^2 a_0/Z$, we can approximate (for n=1)

$$r_1 \approx \frac{1^2 a_0}{Z_1} = \frac{1^2 a_0}{Z-2}$$

- The electrons in the inner shells have large and negative energies. Using Bohr's model, i.e. $E_n = (Z/n)^2(-13.6\ eV)$, we can approximate (for n=1)

$$E_{n=1} \approx (Z-2)^2(-13.6)\ eV$$

- The electrons in the outermost shell have $Z_n \approx n$, so from $r_n = n^2 a_0/Z$, we have $r_n = n a_0$, (overestimates by a factor of 2)

Notice that the multielectronic atom is **not** much greater than hydrogen's atom.

Indeed, the radius of the elements with the highest atomic numbers is roughly $3a_0$.

# Hartree model. Results

- The electrons in the outermost shell have energies comparable to that of the electron in the ground state of hydrogen.

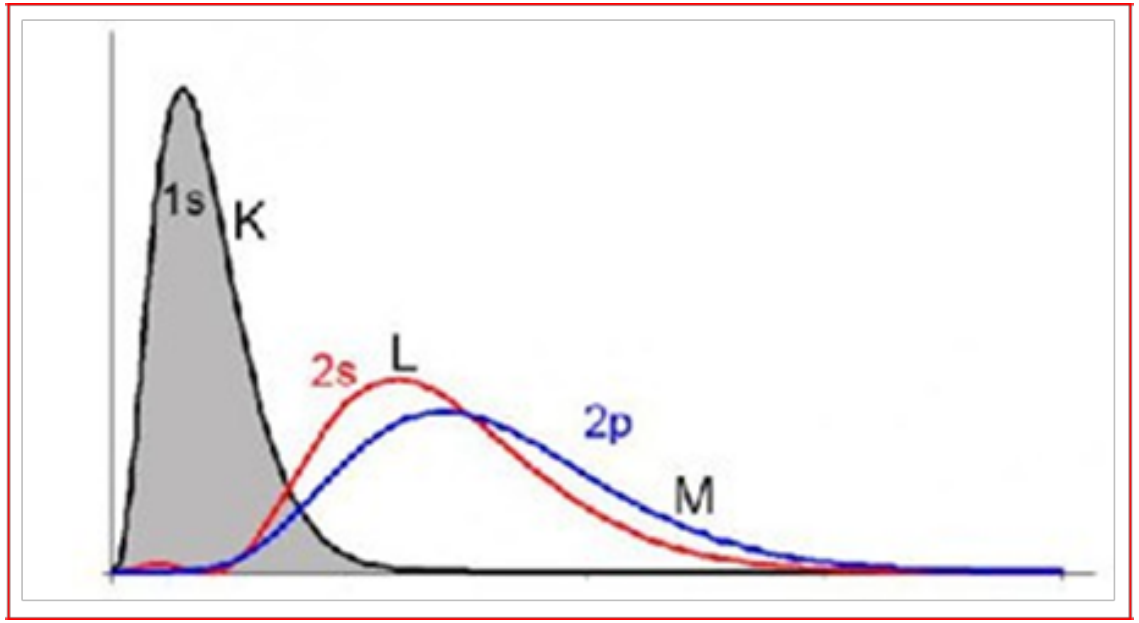
  From $E_n \approx (Z_n^2/n^2)\,(-13.6)$ eV take $Z_n \approx n$ and then $E_n \approx (-13.6)$ eV

- According to Hartree's theory, the energy of an electron depends on ($n, l$), that is, we write $E_{n,l}$... Why?

  This is so because $R_{n,l}(r \to 0) \propto r^l$. Therefore, $\psi^*\psi \propto r^{2l}\ (r \to 0)$

- For fixed $n$, there is a greater probability of finding an $l = 0$ electron in the small $r$ region than that of finding an $l = 1$ electron...(notice that for small $r$, we have $r^0 >> r^2 >> r^4 >>$ ...)

# Hartree model. Results




Radial charge distribution of the kernel and of the valence electron in the 2*s* and 2*p*-states in the Li atom.

# Hartree model. Results

- The electrons in the outermost shell have energies comparable to that of the electron in the ground state of hydrogen.

  From $E_n \approx (Z_n^2/n^2)\,(-13.6)$ eV take $Z_n \approx n$ and then $E_n \approx (-13.6)$ eV

- According to Hartree's theory, the energy of an electron depends on ($n$, $l$), that is, we write $E_{n,l}$... Why?

  This is so because $R_{n,l}(r \to 0) \propto r^l$. Therefore, $\psi^*\psi \propto r^{2l}\ (r \to 0)$

- For fixed $n$, there is a greater probability of finding an $l$ = 0 electron in the small $r$ region than that of finding an $l$ = 1 electron...(notice that for small $r$, we have $r^0 >> r^2 >> r^4 >>$ ...)

  Consequently, the $l$ = 0 electron feels a stronger effective nuclear charge which implies a greater (negative) potential energy, and therefore a smaller total energy.

- Within each shell, we have several subshells, one for each allowed $l$ -value. All electrons in the same subshell have the same quantum numbers (n, l) and, therefore, the same energy (**within Hartree's model!**).

# Hartree-Fock model

A related, but better, approximation takes into account explicitly the antisymmetric character of the many-electron wavefunction, leading to the **HARTREE-FOCK model**, in which the potential felt by the electron in state *i* is:

$$V_i(\vec{r}) = -\frac{Z}{r} + \sum_{j=1}^{N} \int d\vec{r}_2 \frac{\left|\phi_j(\vec{r}_2)\right|^2}{|\vec{r}-\vec{r}_2|} \underbrace{- \delta\left(m_{s_i}, m_{s_j}\right) \sum_{j=1}^{N} \int d\vec{r}_2 \frac{\phi_j^*(\vec{r}_2)\phi_j(\vec{r})\phi_i(\vec{r}_2)}{|\vec{r}-\vec{r}_2|\phi_i(\vec{r})}}_{\text{exchange term}}$$

and (after angular average) along with the equation

$$-\frac{1}{2}\nabla^2\phi_i(r) - \frac{Z}{r}\phi_i(r) + \left(\sum_{j=1}^{N} \int d\vec{r}_2 \frac{\left|\phi_j(\vec{r}_2)\right|^2}{|\vec{r}-\vec{r}_2|}\right)\phi_i(r) - \delta\left(m_{s_i}, m_{s_j}\right) \sum_{j=1}^{N} \int d\vec{r}_2 \frac{\phi_j^*(\vec{r}_2)\phi_j(\vec{r})\phi_i(\vec{r}_2)}{|\vec{r}-\vec{r}_2|} = e_i\phi_i(r)$$

must be solved selfconsistently.

# Hartree-Fock model. Asymptotic behaviour of V(r)

- In the Hartree-Fock eqns., the orbital to be calculated, also appears in the expression for the potential. This means that its resolution is somewhat more complicated than in the case of Hartree eqns.

- Even though $V(r)$ must be computed numerically, in fact it represents the screening of the nuclear charge due to the electronic charge, so it is possible to explain qualitatively its asymptotic behavior.

- For $r \to 0$ there is no screening, and therefore $V(r) \to -Z/r$ (the full nuclear potential). On the other hand, for $r \to \infty$ the electron cloud (except the electron we are considering) screens the whole nuclear charge minus 1, and consequently $V(r) \to -1/r$

- In approximate models this behaviour can in fact be different; for example, in the Hartree model the electron we are considering also belongs to the screening cloud (technically this is called self-interaction) and consequently $V(r)$ goes to zero faster than $-1/r$ (basically exponentially) for large distances.

# Hartree model. Results

Also, even though the eigenvalues $e_{n\ell}$ must be computed numerically, it is possible to explain their dependence on $\ell$, based on the behaviour of both the potential and the radial wavefunctions.

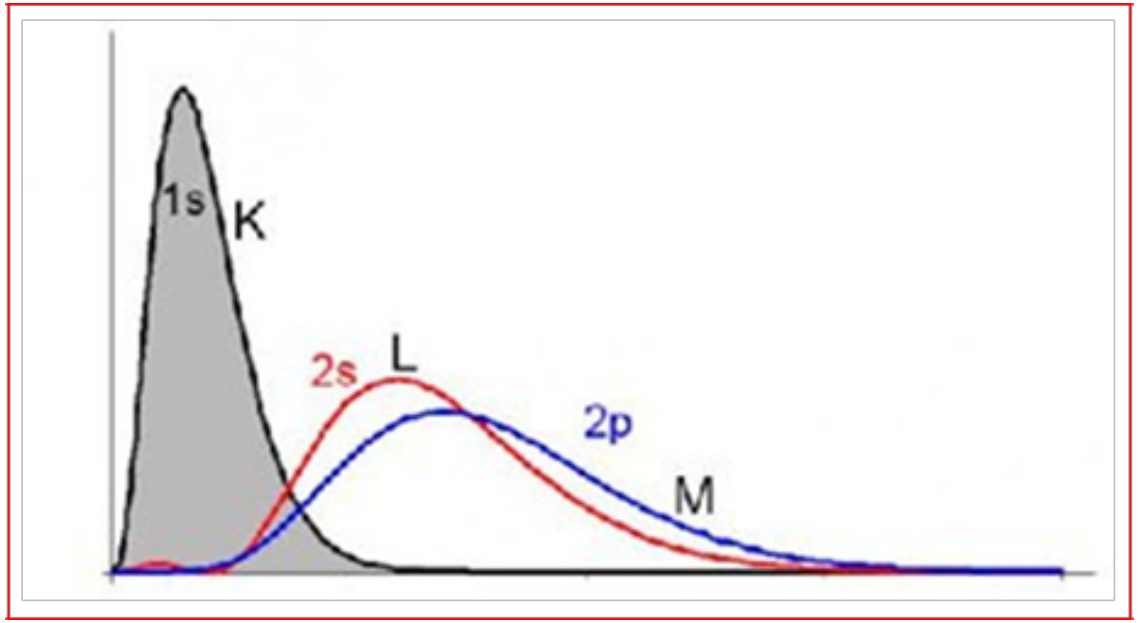
For a given *n*, the radial wavefunctions for small distances behave like $R_{nl}(r) \to r^l$.

So, the smaller $\ell$ is, the more penetrating the wavefunctions are, and consequently they feel more intensely the small *r* dependence of *V* (*r* ), which is more attractive (less screened) as *r* becomes closer to zero. As a consequence, the eigenvalues are smaller (more negative) the smaller the value of $\ell$ is: $e_{n0} < e_{n1} < \cdots < e_{n,n-1}$

*For instance*, $e_{3s} < e_{3p} < e_{3d}$

However, nothing can be said about the ordering for different values of *n*.

# Hartree model. Results




Radial charge distribution of the kernel and of the valence electron in the 2*s* and 2*p*-states in the Li atom.

# Hartree model. Results

All electrons with the same *n* constitute a **shell**. Within a shell, the electrons with the same (*n*, *l* ) constitute a **subshell**.

The electrons in the same subshell ( same (*n*, *l* ) ) are called equivalent electrons.

The state of the atoms is specified by stating the number of equivalent electrons in each subshell. This is called a configuration.

Hartree's model predicts that the energy of a subshell becomes more negative with decreasing values of *n* and decreasing values of *l* .

The subshell 1s has the lowest energy. *n* = 2 has two subshells. The 2s has lower energy than the 2p

In the subshell (n, l) we can allocate 2 (2l+1) electrons. When all these states are occupied, we have a closed shell.

The charge density for a filled subshell is **spherically symmetric**

$$2\sum_{m=-l}^{l}\left|\phi_{n,l,m}(r,\theta,\varphi)\right|^2 = 2\frac{2l+1}{4\pi}R_{n,l}^2(r)$$

# Periodic system

The characteristics of an atom depend on the behaviour of its electrons and the electron's behaviour is specified by four quantum numbers.

But within Hartree's model, only two quantum numbers ($n, l$) are important. Therefore, the atom's state will be characterized by specifying the ($n, l$) quantum numbers of ALL the electrons (**atomic configuration**)

In the ground state, the electrons fill all the subshells in such a way that the total energy is minimized. **They are filled in order of increasing energy**.

In Hartree's model, the energy of a subshell becomes more negative with decreasing values of ($n, l$). The filling goes as:

$$1s,\ 2s,\ 2p,\ 3s,\ 3p,\ 4s,\ 3d,\ 4p,\ 5s,\ 4d,\ 5p,\ 6s,\ 4f,\ 5d,\ 6p,\ \ldots$$

# Periodic system

**The Energy Ordering of the Outer Filled Subshells**

| Quantum Numbers $n, l$ | Designation of Subshell | Capacity of Subshell $2(2l + 1)$ | |
|---|---|---|---|
| — | — | — | |
| — | — | — | |
| 6, 2 | 6*d* | 10 | |
| 5, 3 | 5*f* | 14 | |
| 7, 0 | 7*s* | 2 | |
| 6, 1 | 6*p* | 6 | |
| 5, 2 | 5*d* | 10 | |
| 4, 3 | 4*f* | 14 | |
| 6, 0 | 6*s* | 2 | ↑ |
| 5, 1 | 5*p* | 6 | |
| 4, 2 | 4*d* | 10 | Increasing energy (less negative) |
| 5, 0 | 5*s* | 2 | |
| 4, 1 | 4*p* | 6 | |
| 3, 2 | 3*d* | 10 | |
| 4, 0 | 4*s* | 2 | |
| 3, 1 | 3*p* | 6 | |
| 3, 0 | 3*s* | 2 | |
| 2, 1 | 2*p* | 6 | |
| 2, 0 | 2*s* | 2 | |
| 1, 0 | 1*s* | 2 | ←Lowest energy (most negative) |

# Periodic system

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In the ground state, the electrons fill all the subshells in such a way that the total energy is minimized. **They are filled in order of increasing energy**.

In Hartree's model, the energy of a subshell becomes more negative with decreasing values of ($n$, $l$ ). The filling goes as:

$$1s,\ 2s,\ 2p,\ 3s,\ 3p,\ 4s,\ 3d,\ 4p,\ 5s,\ 4d,\ 5p,\ 6s,\ 4f,\ 5d,\ 6p,\ \ldots$$

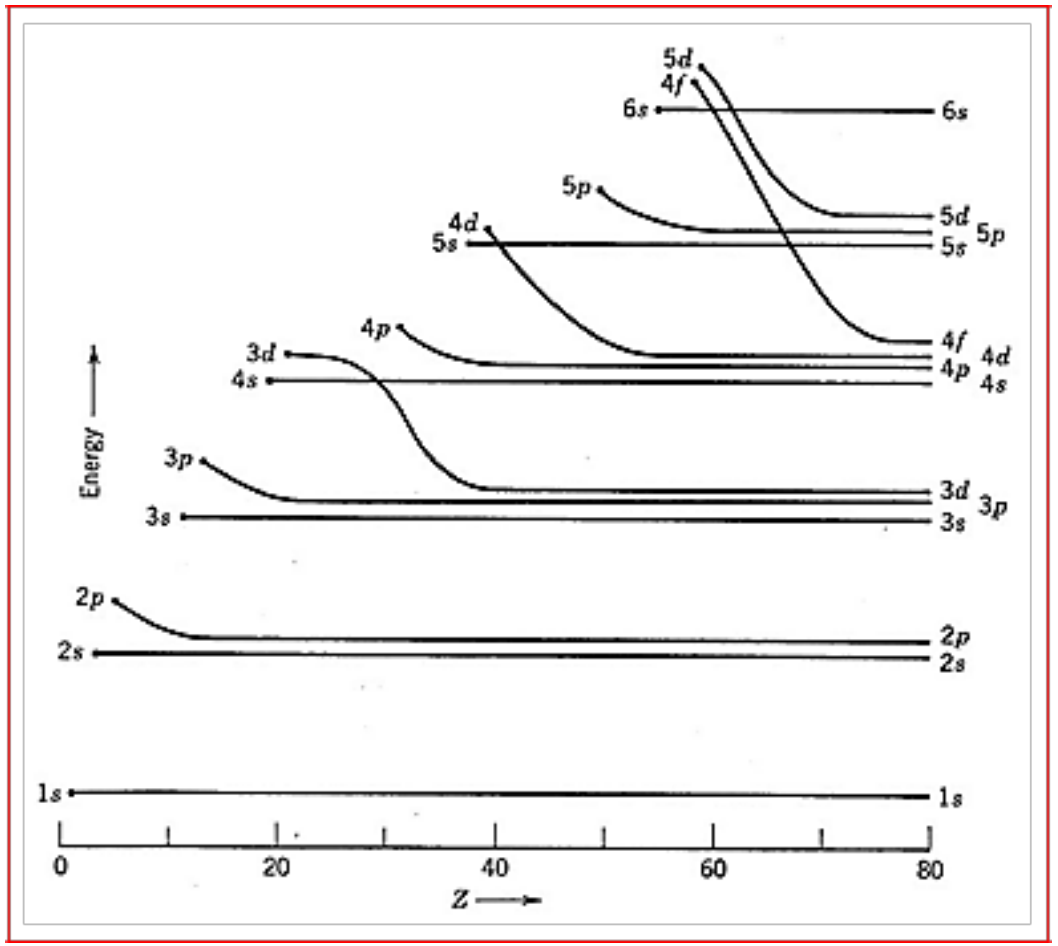
Notice that $E(4s) \leq E(3d)$ because the $l$ -dependence becomes more important than the $n$-dependence. **(It applies to the outer shells)**

This is the case for K and the next few atoms.

BUT for atoms further up in the periodic Table, the ordering is $E(4s) \geq E(3d)$

# Periodic system

Energy ordering of all the subshells in an atom as a function of its atomic number.



# Periodic system

There are some exceptions

$^{23}$V $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^3$

$^{24}$Cr $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1, 3d^5$

$^{43}$Tc $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}, 4p^6, 5s^2, 4d^5$

$^{44}$Ru $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}, 4p^6, 5s^1, 4d^7$

$^{45}$Rh $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}, 4p^6, 5s^1, 4d^8$

$^{46}$Pd $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}, 4p^6, 5s^0, 4d^{10}$

In fact, the energy separation is very small between 4*s* and 3*d* , 5*s* and 4*d* , 6*s* and 5*d* , WHILE they are being filled and sometimes the usual ordering may be reversed. Another example

$^{57}$La $1s^2, 2s^2, 2p^6$ ..... $5s^2, 4d^{10}, 5p^6, 6s^2, 5d^1$

$^{58}$Ce $1s^2, 2s^2, 2p^6$ ..... $5s^2, 4d^{10}, 5p^6, 6s^2, 4f^2$

After filling the 6*s*, we should fill the 4*f* but in some cases the electron goes to the 5*d* . Again this is so because there is a very small energy separation between 6*s* and 5*d* WHILE they are being filled.

Similar arguments apply for the energy separation between 5*d* and 4*f* , 6*d* and 5*f* .

# Periodic system

However, some trends always apply:

Every $(n)p$ subshell has higher energy than the preceeding $(n-1)s$ and $(n-1)d$ subshells, while these are being filled.

Every $(n)s$ subshell has higher energy than the preceeding $(n-1)p$ subshell. There is a large energy difference because they belong to different shells.

Example:

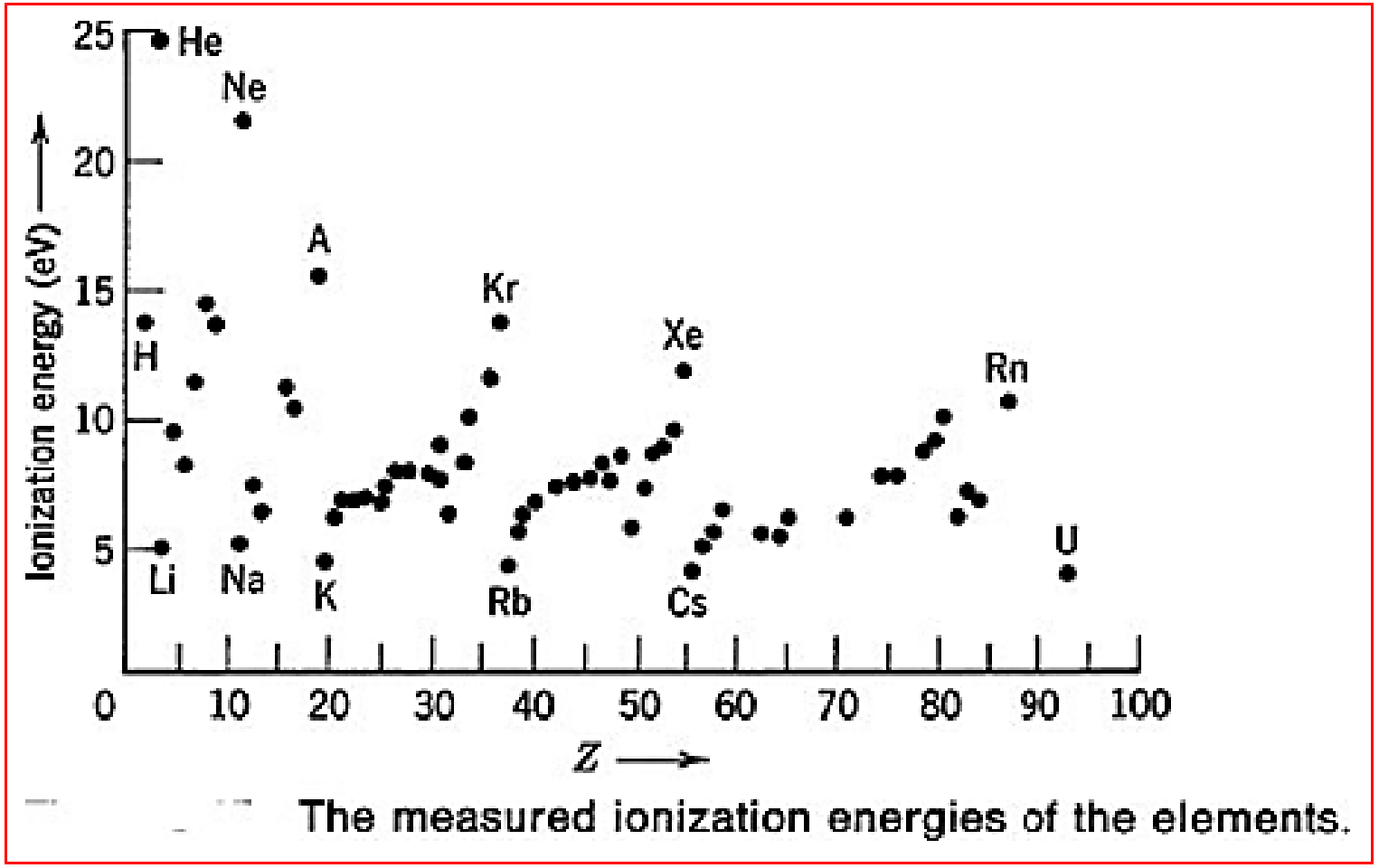
$^{10}Ne$, $^{18}Ar$ , $^{36}Kr$ , $^{54}Xe$ and $^{86}Rn$ have a complete $p$ subshell and their first excited state is very far above the ground state.

Ionization Energy: Total energy of the electron in the highest-energy filled subshell

The values of the ionization energy throughout the Periodic Table, oscillate inside a narrow range (4-10 eV)

It is highest for the noble gases and smallest for the alkali metals. He (≈ 25 eV), Ne (≈ 22 eV), Ar, Kr, Xe (≈ 15 eV)

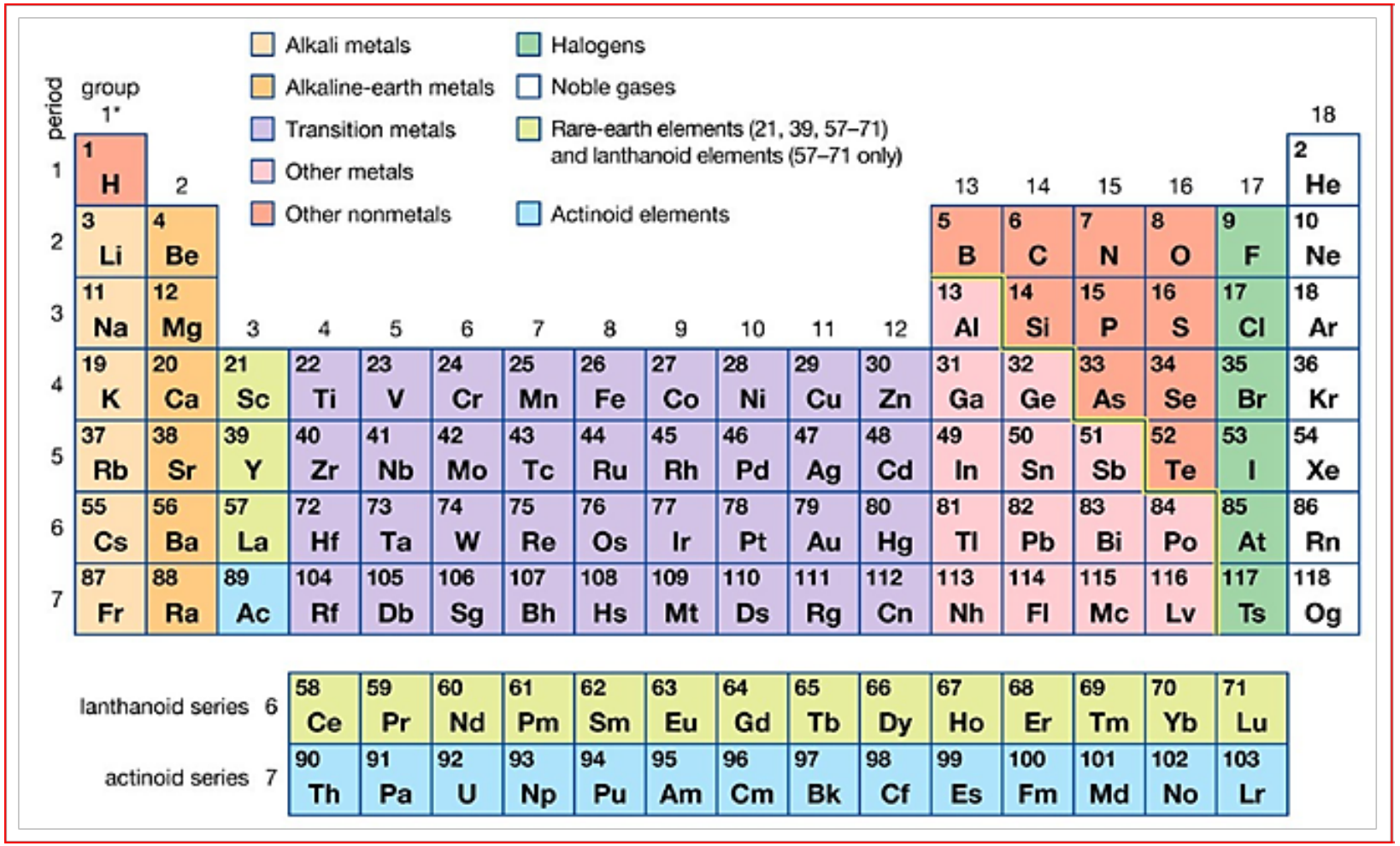
# Periodic system




The measured ionization energies of the elements.

# Periodic system

For the first three rows of the Periodic Table, most properties of elements change uniformly from the alkali to the noble gas



# Periodic system

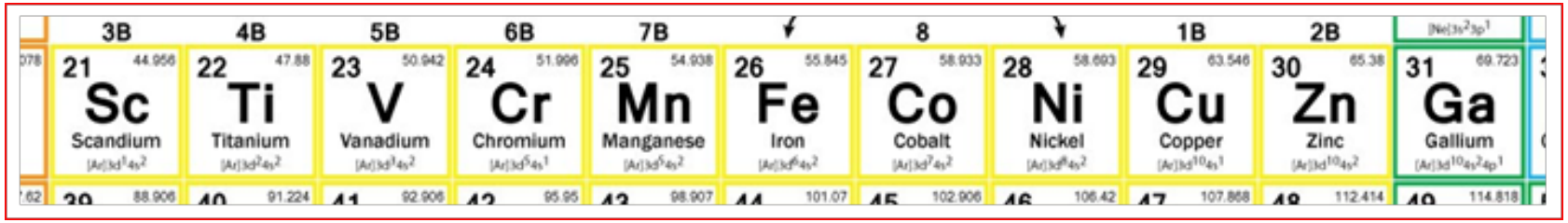
For the first three rows of the Periodic Table, most properties of elements change uniformly from the alkali to the noble gas

Fourth row. Things change somewhat

Elements from $^{21}Sc$ to $^{28}Ni$ are the first transition group with very similar chemical properties. These elements occur during the filling of the 3*d* subshell. (Notice that the 4*s* subshell is already filled, excepting for $^{24}Cr$).

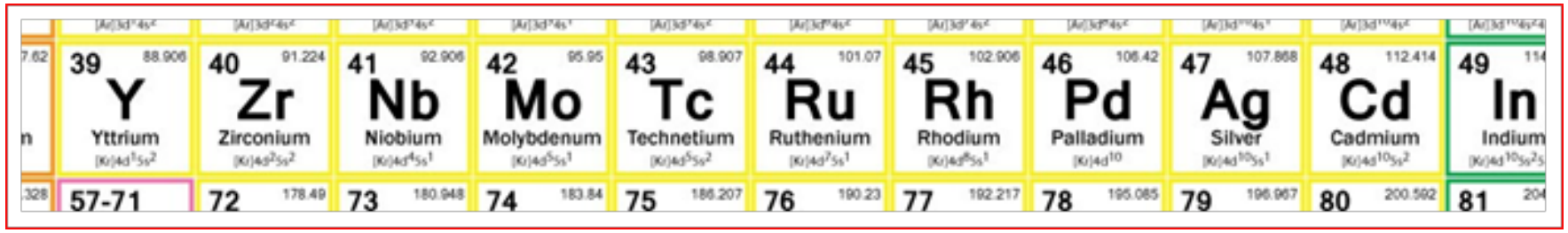
But the radius of the 3*d* << than the radius of the 4*s* and so, the 4*s* subshell shields the 3*d* subshell from external influences; consequently the chemical properties of these elements are very similar.

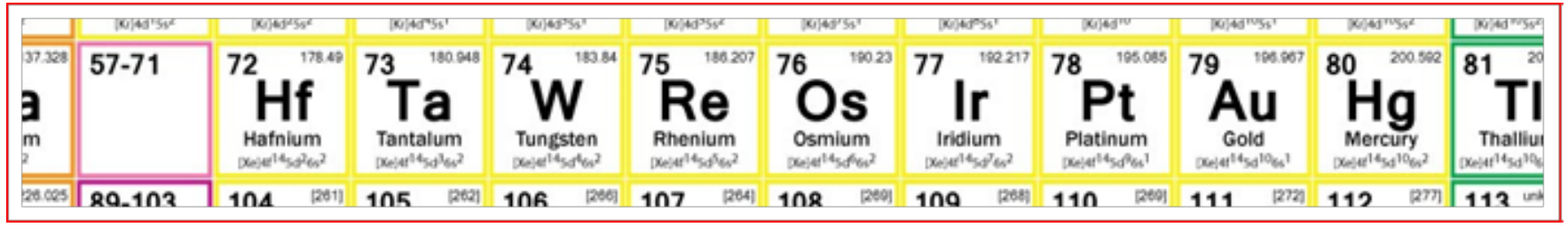
But the chemical properties of $^{29}Cu$ are somewhat different: $4s^1$, $3d^{10}$ $^{30}Zn$ has $4s^2$, $3d^{10}$ (filled subshells) is somewhat more inert.



# Periodic system

Similar arguments apply to the fifth row (filling of the 4*d* subshell) and sixth row (filling of the 5*d* subshell)

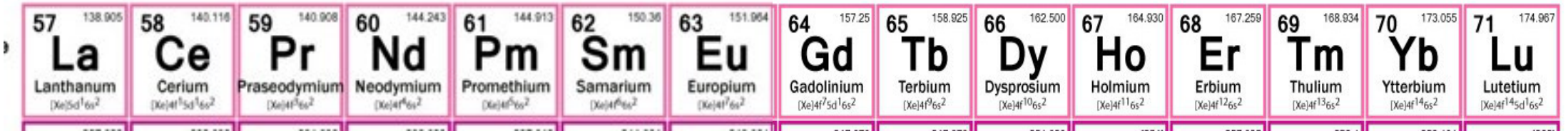


# Periodic system

Similar arguments for the lanthanides and actinides

From $^{58}Ce$ to $^{71}Lu$ there is a filling of the 4*f* subshell. The 6*s* was already filled and it somehow shields the 4*f* electrons from external influences.




From $^{90}Th$ to $^{103}Lr$ there is a filling of the 5*f* subshell. The 7*s* was already filled and it somehow shields the 5*f* electrons.

